Kontrol Sistemleri

Positive feedback root locus, lead/lag compensators

Original notes: Control-Systems-lecture-12.pdf

UYARI Bu belge, el yazısı ders notlarından Claude (Anthropic) yardımıyla dijitale aktarılmıştır. Yardımcı kaynak olarak hazırlanmıştır; hata veya eksik içerebilir. Lütfen orijinal notları ve ders kaynaklarını esas alın.

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orig. p. 1

Positive Feedback architecture

Figure: Two feedback loops drawn as block diagrams. Top: R enters a summing junction marked + (input) and + (feedback input from below); then block k, then block $G(s)$, output $C(s)$; the output is fed back to the junction. Next to it: $k > 0$. Bottom: $R(s)$ enters a summing junction (sign mark at the top hard to read, − at the feedback input); then block k', then block $G(s)$, output $C(s)$; output fed back to the junction. Next to it: $k' < 0$. Red label under the feedback path: "negative feedback".
(in the original notes, p. 1)

$$\frac{C(s)}{R(s)} = \frac{k\,G(s)}{1 - k\,G(s)H(s)}$$

$$1 - k\,G(s)H(s) = 0$$

$$G(s)H(s) = \overbrace{\frac{1}{k}}^{\text{positive}} + 0j$$

$$\angle G(s)H(s) = \tan^{-1}\left(\frac{0}{\varepsilon^{+}}\right) = \pm 360\,i \quad \text{or} \quad \pm 180(2i)$$

$$\boxed{|G(s)H(s)| = \frac{1}{k}}$$

positive feedback

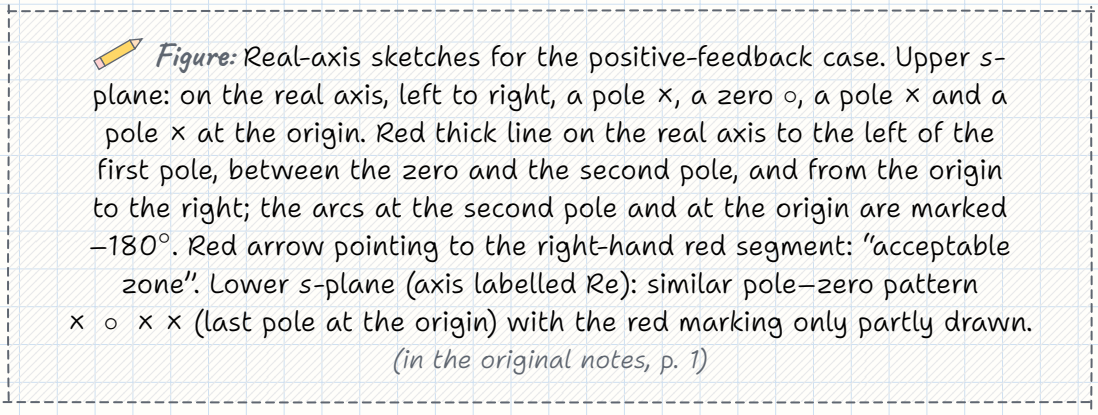


Figure: Real-axis sketches for the positive-feedback case. Upper s-plane: on the real axis, left to right, a pole ×, a zero ○, a pole × and a pole × at the origin. Red thick line on the real axis to the left of the first pole, between the zero and the second pole, and from the origin to the right; the arcs at the second pole and at the origin are marked −180°. Red arrow pointing to the right-hand red segment: "acceptable zone". Lower s-plane (axis labelled Re): similar pole–zero pattern × ○ × × (last pole at the origin) with the red marking only partly drawn.
(in the original notes, p. 1)

(2) Departure angles from *complex* poles
arrival " to *complex* zero

orig. p. 2

departure angle = 360 ?*illegible* + ∑ phase of c.c pole w.r.t. other poles
− ∑ phase of c.c pole w.r.t. other zeros

(3) Asymptote angles

$$\frac{\pm 360\, i}{n-m}$$

Example

$G(s) = \dfrac{k(s+2)}{(s+3)(s^2+2s+2)}$
Positive feedback system

$$z_1 = -2$$
$$p_1 = -3$$
$$p_{2,3} = -1 \pm j$$

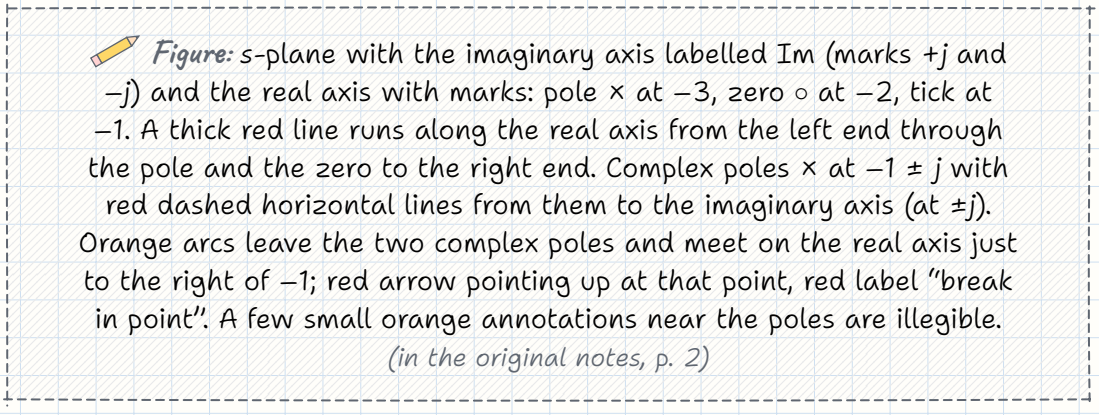


Figure: s-plane with the imaginary axis labelled Im (marks +j and −j) and the real axis with marks: pole × at −3, zero ○ at −2, tick at −1. A thick red line runs along the real axis from the left end through the pole and the zero to the right end. Complex poles × at −1 ± j with red dashed horizontal lines from them to the imaginary axis (at ±j). Orange arcs leave the two complex poles and meet on the real axis just to the right of −1; red arrow pointing up at that point, red label "break in point". A few small orange annotations near the poles are illegible.
(in the original notes, p. 2)

branches: 3
$n - m = 3 - 1 = 2$
angle of asymptotes: $\dfrac{\pm 360\, i}{n-m} = \pm 180\, i$

for $s = -1 + j$

$$\theta_1 = 0° - 27° - 90° + 45° = -72°$$

for $s = -1 - j$

$$\theta_2 = 0° + 27° + 90° - 45° = +72°$$

To find characteristic equation

$$1 + \frac{k(s+2)}{(s+3)(s^2+2s+2)} = 0$$

$$(s+3)(s^2+2s+2) + k(s+2) = 0$$

$$k = -\frac{(s+3)(s^2+2s+2)}{(s+2)}$$

$$\frac{dk}{ds} = 0 \Rightarrow 2s^3 + 11s^2 + 20s + 10 = 0$$

$$\Rightarrow 2(s+0{,}8)(s+2{,}35+j0{,}77)(s+2{,}35-j0{,}77) = 0$$

orig. p. 3

$\Rightarrow s = -0{,}8$ break in

if its negative feedback

> *Figure:* s-plane: real axis with pole × at −3, zero ○ at −2 and mark −1; complex poles × at $-1 \pm j$ (imaginary axis to the right). Red arrow on the real axis from the pole at −3 to the zero at −2. Red dashed vertical line (asymptote) crossing the real axis at about −1,5, with a small red label at its lower end (probably −1,5); red curved branches come down from above to the upper complex pole and up from below to the lower complex pole, following the dashed line.
> *(in the original notes, p. 3)*

$$\text{centroid} = \frac{\sum \text{real part poles} - \sum \text{real part of zeroes}}{n-m} = -1{,}5$$

Example

Negative feedback

> *Figure:* Two s-plane sketches side by side. Left: two real poles × left of the imaginary axis; thick red segment between the two poles and a thick red vertical line through the middle of that segment, extending up and down. Right: zero ○ on the real axis, complex poles × above and below the real axis to the right of the zero (imaginary axis further right); a red arc leaves the upper pole, curves to the left, crosses the real axis to the left of the zero and returns to the lower pole (arrows on the arc); on the real axis a red arrow points left from the crossing point and a red segment with arrow leads right from the crossing point to the zero.
> *(in the original notes, p. 3)*

Pozitive feedback

[0] Claude notu: pozitif geri besleme için karakteristik denklem muhtemelen $1 - kG(s) = 0$ olmalı; $dk/ds = 0$ kökleri bundan etkilenmez.

Figure: Two s-plane sketches side by side. Left: two real poles × left of the imaginary axis; thick red line on the real axis to the left of the left pole (arrow pointing left) and from the right pole to the right through the imaginary axis to the right end; nothing between the two poles. Right: zero ○ on the real axis, complex poles × above and below the real axis to the right of the zero, close to the imaginary axis; red arcs from the poles curve to the real axis just left of the imaginary axis; thick red line on the real axis from the zero to the right with arrows pointing left (towards the zero) and right.
(in the original notes, p. 3)

Negative feedback

Figure: Root-locus sketch with real-axis marks −5, −4, −2,5, −1, 0 (poles × at −5, −4, −1 and 0). Two red dashed asymptote lines cross at −2,5 (one rising to the right, one falling to the right). Thick red segments on the real axis between −5 and −4 and between −1 and 0; red curved branches with arrows leave these segments, the left pair bending towards the left asymptote directions and the right pair towards the right asymptote directions.
(in the original notes, p. 3)

$$\frac{\pm 180(2i+1)}{n-m}$$

$$\pm 45(2i+1)$$

$$\pm 45 \qquad \pm 135$$

centroid = −2,5

orig. p. 4

Pozitif f? *illegible*

Figure: s-plane with axes labelled Im (top) and Re (right end of the real axis). Four poles × on the real axis, the last one at the origin and the other three at negative real values. Thick red line on the real axis to the left of the leftmost pole (arrow pointing left), between the second and third poles (arrows pointing towards each other) and from the origin to the right end. At the point between the second and third poles a red vertical line with arrows pointing up and down.
(in the original notes, p. 4)

$$\frac{\pm 360\, i}{4} = \pm 90\, i$$

Compensator design

Figure: s-plane with a pole × on the negative real axis and a pole × at the origin; thick red segment between them with an arrow pointing right. Red branches: an arc on the left (from the left pole up-left and down-left, arrows on the ends) and a vertical arc just left of the origin (arrow pointing up at the top end, down at the bottom end). Two orange crosses mark the "desired dominant poles" (one above, one below the real axis, left of the imaginary axis, not on the locus); a black dashed line from the upper one towards the origin with the angle θ marked at its end.
(in the original notes, p. 4)

specific "k" which renders the system unstable

4 seperate roots of ch.e.q.

Figure: s-plane. Thick red segment on the negative real axis ending at a pole × labelled $-p$; zero ○ labelled -2 with a red arrow pointing left; pole × at the origin. Red arcs leave the pole at $-p$ to the upper left and lower left (arrows), black arcs leave the origin to the upper right and lower right (lower one with a red arrow). Label to the right of the axis: $k = 0$.
(in the original notes, p. 4)

$$1 + \frac{k(s+2)}{s^3(s+p)^3}$$

$\theta = \cos^{-1}\zeta$ damping ratio

Figure: s-plane. Real-axis marks, left to right: red ○ at the far left labelled $-z_2$; blue × labelled $-p_3$; ○ labelled $-z_1$; × labelled $-p_2$; × labelled p_1 (at the origin, on the imaginary axis). A test point s above the axis (label hard to read) is joined by a blue dash-dotted line to the blue pole $-p_3$, with the angle 10° marked there, and by red lines to $-z_1$, $-p_2$ and p_1, with the angles ϕ_1, θ_2, θ_1 marked at those points.
(in the original notes, p. 4)

↘ $\frac{1}{s+p_3}$ → delayed integrator

$$F(s) = \frac{s+a}{s+b} = \frac{R e^{i\phi}}{M e^{i\theta}} = \left(\frac{R}{M}\right) e^{i(\phi-\theta)}$$

Phase of ?s points $\beta = \phi_1 - \theta_1 - \theta_2$

the original OLTF $\frac{s+z_1}{(s+p_1)(s+p_2)}$

if $\beta = 190°$ Negative $\frac{-10}{180}$ → to make it negative p_3 angle have to be bigger

?PI $\frac{s+z_2}{s+p_3} \simeq \frac{1+z_2}{\lim_{p_3 \to 0} s}$

orig. p. 5

if $\beta = 150°$ $\frac{30°}{180°}$

Figure: s-plane. Real-axis marks, left to right: red ○ labelled $-z_2$ (far left); ○ labelled $-z_1$; × labelled $-p_2$; × at the origin labelled p_1. An orange test point s above the axis is joined by a red dashed line to $-z_2$ (angle 30° marked in red) and by black dashed lines to $-z_1$, $-p_2$ and p_1, with the angles ϕ, θ_2, θ_1 marked at those points.
(in the original notes, p. 5)

$$\underbrace{(s+z_2)}_{\text{derivative operator}} \quad \frac{(s+z_1)}{(s+p_1)(s+p_2)}$$

Its unuseless itself so we have to add a pole

Figure: Same sketch with an extra red pole × at the far left labelled $-p_3$; the red zero ○ labelled $-z_2$ now lies to the right of it, with a red arrow pointing at it and the red label "New z_2". Red dashed lines join the orange test point s to $-p_3$ and $-z_2$, black dashed lines join it to $-z_1$, $-p_2$ and p_1; angles ϕ, θ_2, θ_1 marked.
(in the original notes, p. 5)

$$\frac{s+z_2}{s+p_3} \underset{\lim z_2 \to 0}{=} \frac{s}{s+p_3} = 1 - \frac{p_3}{s+p_3}$$

?? WTF

Adding poles

Figure: Four root-locus sketches (three in the first row, one in the second). (a) One pole × labelled p_1 on the negative real axis near the imaginary axis; red thick line on the real axis from the left end to the pole. (b) Two poles × (p_1 the right one); red thick segment between them and a red vertical line through its midpoint. (c) Three poles × (p_1 the rightmost); red thick line from the left end to the leftmost pole, red dashed lines from the middle pole going up-right and down-right, and red curved branches that leave between the two right poles and bend into the right half-plane (arrows pointing right). (d) Four poles × on the negative real axis in two pairs; red thick segments between poles 1–2 and 3–4; red arcs: the left pair bends to the left (up and down), the right pair bends to the right across the imaginary axis.

(in the original notes, p. 5)

by adding poles the locus plot move to right, systems get more unstable. (lag compensator)

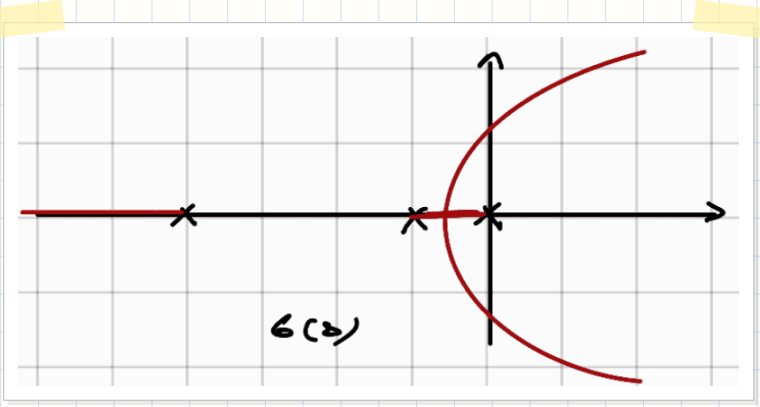
orig. p. 6

System get more unstable - slower! ($t_s \uparrow$)

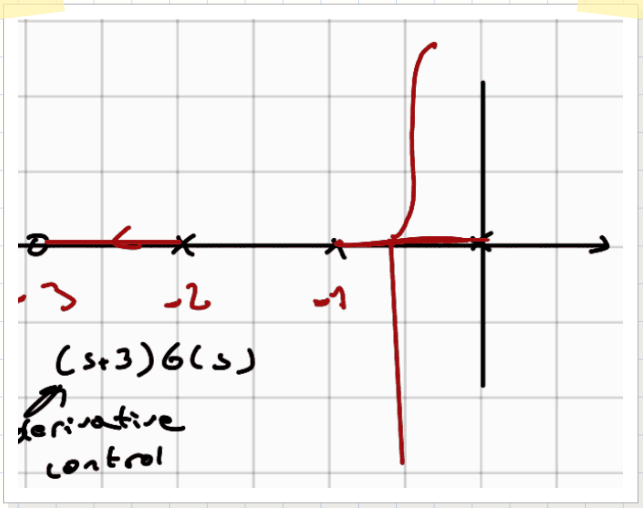
More accurate by adding integrator

lag compensation, The output angle is less than input

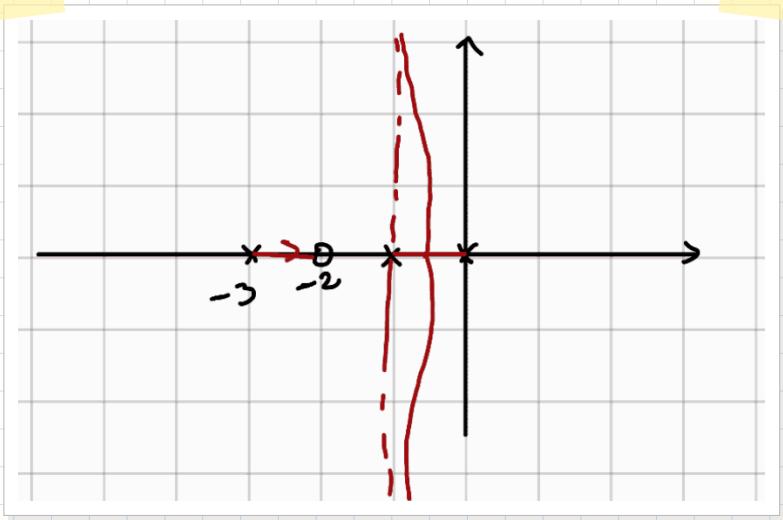
zero placement



original drawing, p. 6

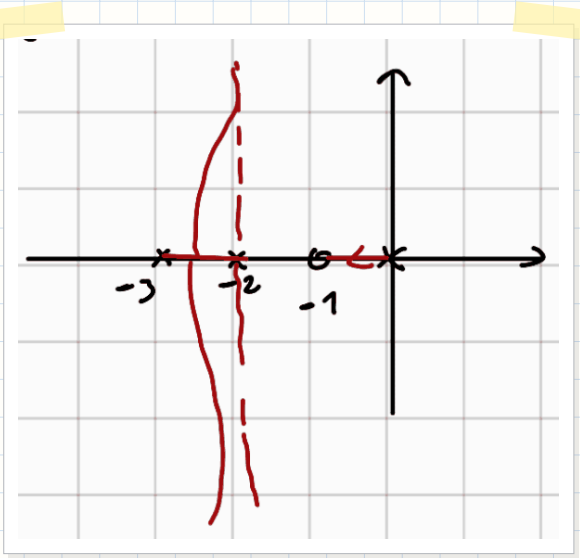


original drawing, p. 6



original drawing, p. 6

$$\frac{-1-3-(-2)}{2} = \frac{-4+2}{2} = -1$$



original drawing, p. 6

pulls the root locus plot to the left

→ more stable

→ faster response ($t_s \downarrow$)

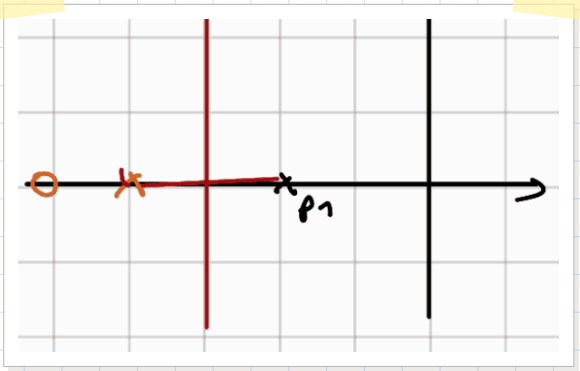
more sensitive to noise

The output angle is more than the input angle

its leading output

$$r(t) = \sin \omega t \quad \rightarrow \quad y(t) = \sin(\omega t + \phi)$$

orig. p. 7

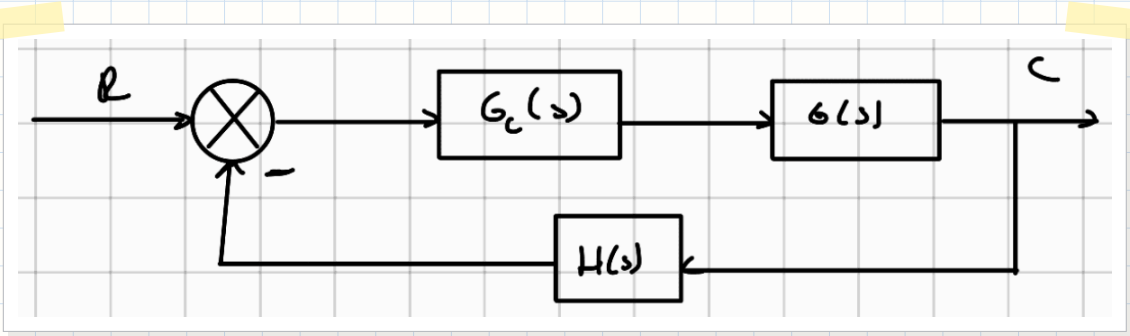


original drawing, p. 7

lag compensation ? $z > p > 0$

lead laging compensator

Series/parallel compensation



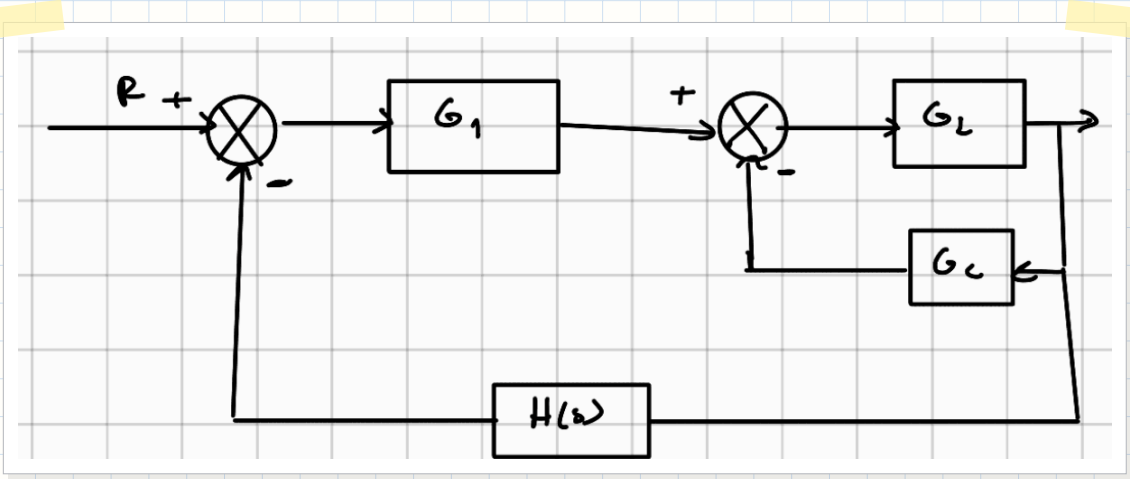
original drawing, p. 7

$$\frac{C(s)}{R(s)} = \frac{G_c G}{1 + G_c G H}$$

$$1 + G_c G H = 0 \Rightarrow G_c G H = -1$$

$\angle G_c G H$ = gives the root locus

parallel



original drawing, p. 7

$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_2 G_c + G_1 G_2 H} \longrightarrow 1 + G_c \frac{G_2}{1 + G_1 G_2 H} = 0$$

$$1 + G_2 G_c + G_1 G_2 H = 0 \Rightarrow 1 + G_c G_p = 0$$

$$\times \frac{1}{1 + G_1 G_2 H}$$

$$= \frac{1 + G_1 G_2 H + G_2 G_c}{1 + G_1 G_2 H}$$

orig. p. 8

Lead compensator: zero is closer to jω axis

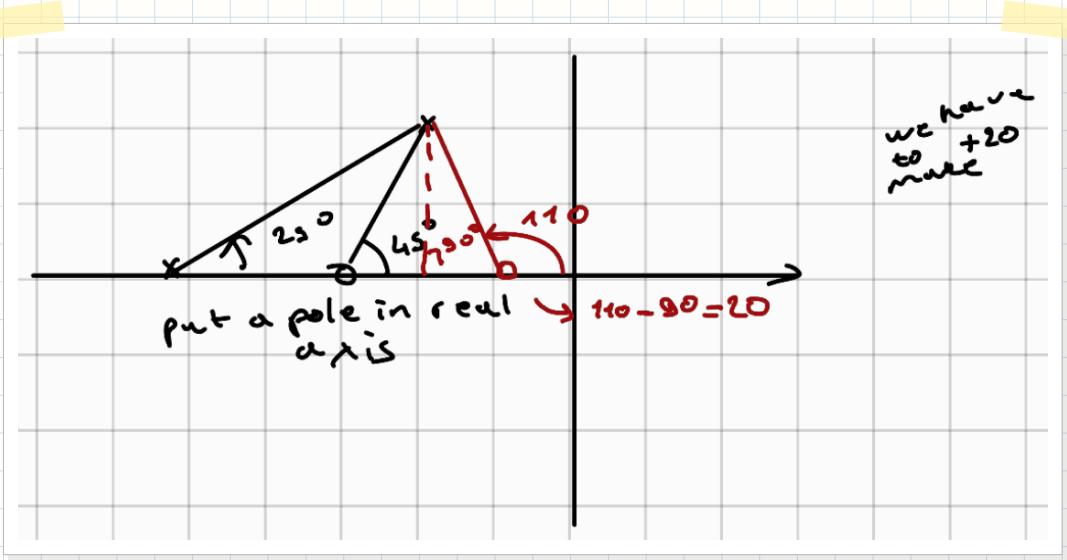
$$G_c(s) = k_c \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} \rightarrow \text{specs} \qquad 0 < \alpha < 1$$

$$k_c \frac{s+z}{s+p}$$

if $\alpha > 1 \Rightarrow$ lag compensator

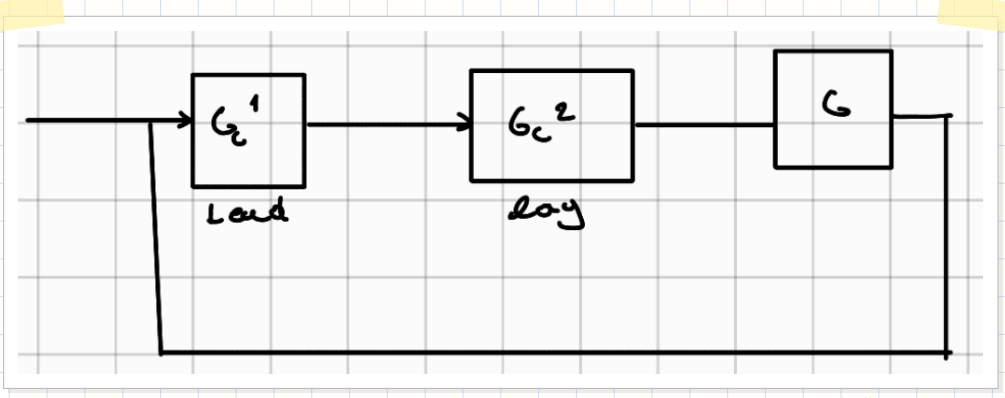
$$G_c(s) = k_c \frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \qquad \beta > 1$$

$$\frac{1}{T} = 10 \qquad \begin{cases} z = -10 \\ p = -\frac{10}{\alpha} = -20 \end{cases} \qquad \alpha = \frac{1}{2}$$



original drawing, p. 8

Lead lag compensator



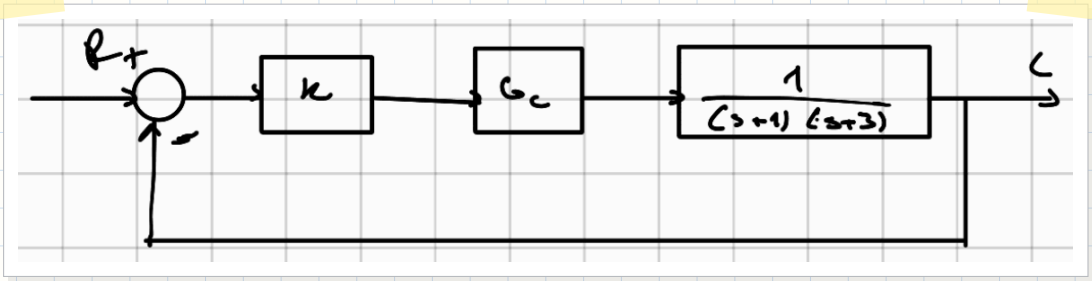
original drawing, p. 8

$$\left(k_c \frac{s + \frac{1}{T_1}}{s + \frac{1}{\alpha T_1}}\right)\left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\beta T_2}}\right)$$

? 5 unknowns to be design

$0 < \alpha < 1 \qquad \beta > 1$

Ex:



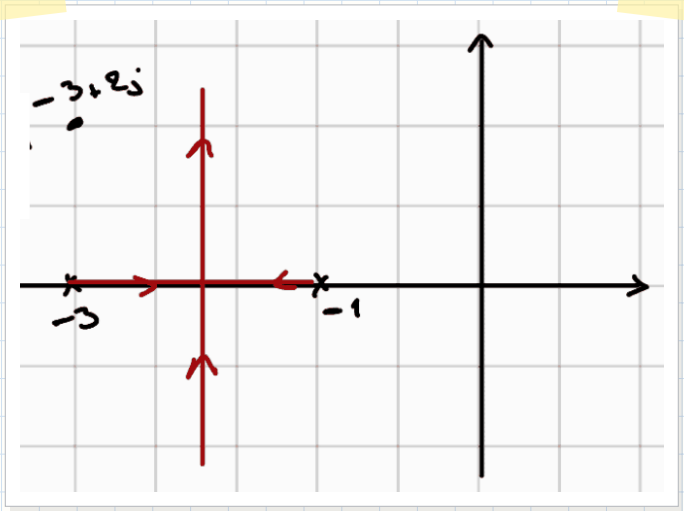
original drawing, p. 8

orig. p. 9

Dominant pole = $-3 + 2j$ → lead compensator

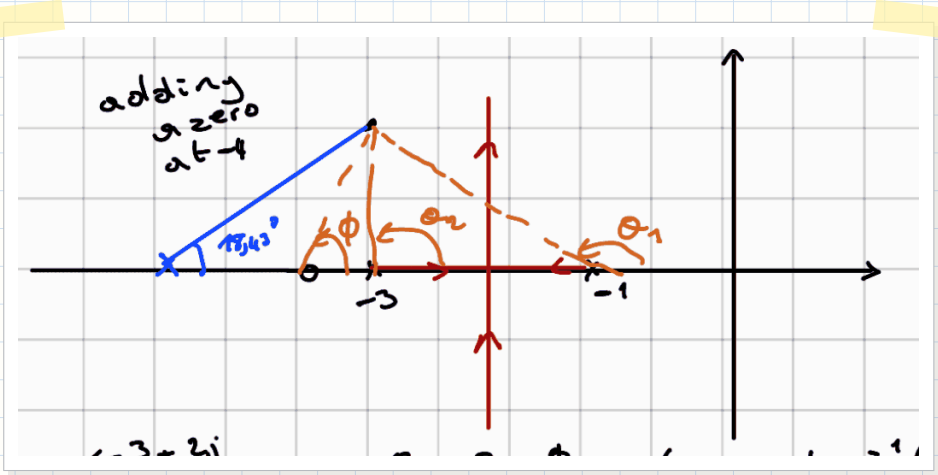
e_{ss} ← Steady state error = 10% (step input) → lag compensator

$G_c = 1$



original drawing, p. 9

$$G_c = \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}}$$



original drawing, p. 9

$$\angle(-3 + 2j) = -\theta_1 - \theta_2 + \phi = -\left(180 - \tan^{-1}\left(\frac{2}{2}\right)\right) - 90 + \tan^{-1}\left(\frac{2}{1}\right)$$

$$= -135 - 90 + 63{,}43^\circ = -161{,}56^\circ$$

phase deficiency $= -135 - 90 + 63{,}43^\circ = -161{,}56^\circ$

So compensator $G_c(s) = \dfrac{s+4}{s+9}$

$$E_{ss} = \frac{1}{1+k_p} \rightarrow \text{static position error constant}$$

$$k_p = \lim_{s\to 0} G(s)$$

$$E_{ss} = \lim_{s\to 0} \frac{sR(s)}{1+G(s)} = \lim_{s\to 0} \frac{1}{1+G(s)} = \frac{1}{1+\lim_{s\to 0} G(s)} = \frac{1}{1+k_p}$$

orig. p. 10

$$G(s) = k_c \left(\frac{s+4}{s+9}\right)\left(\frac{1}{(s+1)(s+3)}\right) \quad \text{compensator system}$$

$$E_{ss} = 0{,}1 = \frac{1}{1+k_p} \Rightarrow 1 + k_p = 10 \Rightarrow k_p = 9$$

$$9 = k_p = \lim_{s\to 0} G(s) = k_c \left(\frac{4}{9}\right)\left(\frac{1}{1\times 3}\right)$$

$$k_c \left(\frac{4}{27}\right) = 9$$

$$\Rightarrow k_c = \frac{9 \times 27}{4}$$

$$G(s) = \left| k \frac{s+4}{s+9} \frac{1}{(s+1)(s+3)} \right| = 1$$

k give dominant pole